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Literature Review First Draft

Introduction

Over the past few decades, science has advanced so far that the community no longer has to make speculations about Mars' habitability. Now, it is grounded in real, detailed, data-driven studies of Martian geochemistry, atmospheric evolution, and biosignature preservation. This literature review is an interdisciplinary discussion of Martian habitability based on a number of different approaches, including alternative biochemistry, geological evidence, environmental conditions, and more. Although the conclusion is that surface life on Mars no longer exists, there is strong evidence pointing to the fact that it could have existed at some point and clues in the form of biosignatures might still remain.

Defining Habitability and Biosignatures

Understanding past life on Mars starts with defining what "habitability" means and how biosignatures can be identified. Apai et al. (2025) argue that habitability should be assessed in flexible ways that can change based on what planet or model of interest. Similarly, Schwieterman & Leung (2024) emphasize that "habitable" conditions will not always look like what is seen to the familiar conditions on Earth. Both of these sources discuss the general research trend of shifting away from clear signs of life to biosignatures and environments where they can be preserved. This is particularly important when it comes to Mars. Although surface life can no longer exist, conditions on Mars do have the ability to preserve signs of past life. Surface conditions, rather than atmospheric climate conditions, are emphasized. Mars' thin atmosphere limits the detection of biosignatures through atmospheric measurements. Geological and geochemical evidence from the surface is therefore much more reliable. As a result, astrobiologists focus on identifying specific environments, such as ancient lakebeds and mineral deposits, where biosignatures are more likely to be preserved over billions of years depending on the age of the rocks.

Atmospheric Evolution and Loss of Surface Habitability

The evolution of Mars' atmosphere is critical in understanding its habitability. Jakosky (2021) discusses how atmospheric loss was driven by solar wind, and then propelled by thermal escape. These factors led to a significant reduction in atmospheric pressure and eliminated the stability of surface water. Cary et al. (2024) furthers this by adding that these changes created increasingly hostile surface conditions, which includes increased radiation exposure and chemical instability. Early Mars, particularly during the Noachian period (4.1 to 3.7 billion years ago), may have supported stable surface liquid water, but as Mars lost its atmosphere, long-term surface habitability declined rapidly. This led into a shift in research focus towards subsurface environments, where conditions could have remained stable even while conditions on the surface changed drastically.

Prebiotic Chemistry and Amino Acid Formation

The availability of organic molecules plays a key role in determining where and how life could emerge. Takeuchi et al. (2020) shows that the building blocks of life, specifically amino acids, can be formed via pathways that are not aligned with the traditional theory of amino acid catalysis on Earth. Specifically, they experimented with the concept of shock-induced catalysis and demonstrated that meteorite impacts on Noachian Mars could have provided the energy needed to perform chemical reactions. Mayer-Bacon et al. (2022) suggests that early amino acid sets here on Earth were non-random and had chemical significance in why they went on to be used in life as we know it. These findings suggest that amino acid availability on Mars may have been spatially variable, but chemically functional. Expanding on this idea, Brown et al. (2026) proved that alternative amino acid alphabets can form stable peptide structures, which indicates that life on Mars may not have relied on the same biochemical processes as life on Earth. But in contrast to these ideas, Cary et al. (2024) point out that environmental constraints could have prevented higher-complexity biochemical structures from forming. Iron-rich conditions on Mars have destabilizing effects on lipid membranes, breaking apart protocells. Together, these studies show tension in the literature in the fact that although organic molecules may have been readily available, environmental conditions that could have supported stable biological systems remain uncertain.

Geological Evidence of Past Water and Biosignature Preservation

Geological studies and surveys provide strong evidence that surface water, although not present today, likely existed on Mars in the past. Ye et al. (2019) and Phillips et al. (2023) both point towards environments such as chloride deposits and salt-encrusted basins as indicators of past hydrology. Evaporative minerals, especially salts, can trap organic molecules and microbial structures within their crystal matrices as they form. As water evaporates, these minerals precipitate and encase organic matter within them. Preserved organic material can serve as powerful biosignatures if Martian mineral samples can be collected. Research focused on Jezero Crater, the landing site of the Mars 2020 Perseverance Rover and believed to be an ancient lakebed, strengthens the case for past surface habitability. Zastrow & Gloch (2021) identified that different carbonate-rich units show different periods of hydrological activity in the crater. In addition to this, Beck et al. (2025) found through analysis of hyperspectral imagery silica-rich deposits like quartz that are associated with hydrothermal systems. Kalucha et al. (2024) also added that different types of mineral concretions found on Earth formed by groundwater activity also can be found in Jezero. These studies support the past presence of a variety of different water-rich environments on Mars, including lakes, hydrothermal systems, and groundwater systems. Environments such as these not only could have hosted life, but also potentially contain evidence of ancient Martian life as well.

Subsurface Habitability and Alternative Metabolisms

As surface conditions rapidly declined and the Martian surface became uninhabitable, subsurface locations likely would have become the primary habitable zones. Cary et al. (2024) and Cowan et al. (2022) both discuss and emphasize the importance of subsurface conditions as important in the formation and sustainability of life. Cowan et al. (2022) interestingly suggests that Martian organisms could survive on metabolizing gases from the atmosphere, termed atmospheric chemotrophy, from under the surface or in low-nutrient environments. When these findings are combined with the geological evidence for groundwater and hydrothermal activity, the idea that Mars can support subsurface ecosystems becomes more compelling and realistic.

Biosignature Detection Methods

The instruments used for biosignature detection are as important as the biosignatures themselves. Foing et al. (2011) emphasize the importance of combining remote sensing methods with that of *in situ* analyses. This source also goes into the idea of using analogue sites on Earth that mimic extraterrestrial sites such as Mars to develop and test instruments before using them in planetary exploration missions. Rull et al. (2017) focuses specifically on Raman spectroscopy as key to Mars sample analysis. Raman spectroscopy measures molecular vibrations and can provide insights into both mineral phases and composition, as well as identifying organic compounds within a mineral sample. Combining these two methods is incredibly important in interpreting Martian geology and potential biosignatures. Remote sensing provides the spatial location of potential biosignature-preserving environments, while *in situ* analyses can confirm the presence of organic matter. However, the effectiveness of these methods depends on the instruments' ability to recognize unfamiliar biochemical patterns. It is uncertain that Martian life would have the same biochemistry and life on Earth, as Brown et al. (2026) showed that it is possible for alternative alphabets of building blocks to form stable structures. Most instrumentation methods are built to identify Earth biochemistry and may have a challenge if Martian biology does not closely mimic that of terrestrial biological systems. This fact points to a need for more flexible instrumentation and analysis methods that account for more biochemical possibilities outside of the ones we are aware of.

Gaps in the Literature

Despite significant scientific advancements when it comes to studying past life on Mars, many gaps still exist in the literature. Firstly, there is limited research that connects the availability of biochemical monomers, such as amino acids, to specific geological environments. While many of the studies discussed in this literature review show evidence that amino acids can form under prebiotic conditions, very little is correlated with where such conditions could be concentrated or preserved on the surface on Mars spatially. Understanding the spatial distribution of organic compounds and their geological record is crucial, as localized environmental factors such as pH, mineral composition, and presence of hydrological activity would directly impact both their formation and long-term preservation. Without this connection,

the relationship between prebiotic chemistry and the actual Martian landscape remains largely theoretical rather than observational.

Many current biosignature detections are designed with Earth-based biological assumptions in mind. Although methods that use this assumption are effective in detecting familiar organic chemistry, they may overlook alternative biochemical systems that do not follow the same patterns as Earth life does. As Brown et. al (2026) demonstrated, it is very possible for non-canonical amino acids to form stable secondary structures that could very well be integrated into stable tertiary protein structure. This creates a limitation in current detection techniques and instrumentation, as they would fail to recognize biosignatures that fall outside of the expected parameters.

Finally, although regions such as Jezero Crater and Gale Crater have been extensively studied by both remote sensing analyses and on-the-ground rover missions, there is still a lack of analyses that combine geochemistry, mineralogy, and spatial data. Many studies focus on the aspects separately, but I found no evidence of studies that synthesize datasets to identify the most promising locations for biosignature preservation. This suggests the need for an interdisciplinary approach that combines spatial analysis skills with the ability to understand the biochemical record to better both interpret and visualize the relationships between geological surface features and biochemical signatures.

Project Direction

The aforementioned gaps in the literature directly inform the direction that I want to take with my capstone project. My project will aim to connect prebiotic chemistry to the geological context of the Martian surface by focusing on the spatial distribution of environments that could support both the formation and preservation of amino acids and simple protein structures. Specifically, my project will use Jezero Crater as the area of interest. By integrating remote sensing data with spatial analysis techniques, my project will seek to provide a more targeted approach to identifying where evidence of past life is most likely to be found. Additionally, this project will consider the possibility of alternative biochemical systems when evaluating potential biosignatures, expanding beyond strictly Earth-based assumptions. Through this approach, my research aims to contribute to a more comprehensive understanding of how and where life could have existed on Mars.

Annotated Bibliography